

# Multi-Timescale Volatility Forecasting: Transferring a Dual-Resolution TCNN-LSTM from Human Activity Recognition

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## Abstract

Realized-volatility forecasting and the heterogeneous autoregressive (HAR-RV) model of Corsi [2] rest on a multi-timescale prior: volatility is best predicted by combining information aggregated over a day, a week, and a month. We observe that this prior is structurally identical to a dual-resolution hybrid TCNN-LSTM that we recently co-proposed for human activity recognition [8], which fuses a short-context and a long-context branch. This project transfers that architecture to one-day-ahead volatility forecasting for the S&P 500 and compares it against GARCH(1,1) and HAR-RV using RMSE, the QLIKE loss, and Diebold-Mariano tests. On the 2021–2026 test set the transferred model attains the lowest squared-error loss and significantly improves on HAR-RV (Diebold-Mariano  $p = 0.013$ ); a five-seed robustness sweep confirms the improvement is stable, and a branch ablation confirms the dual-resolution design contributes. The gain is small in absolute terms because the daily volatility proxy is intrinsically noisy, which we analyze explicitly. The contribution is a clear logical link from a known financial inductive bias to a learned architecture that respects it.

## 1 Introduction and Motivation

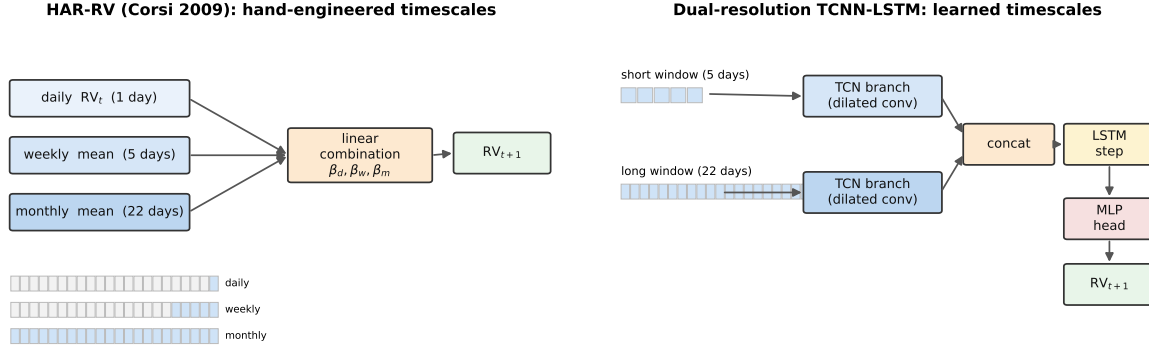
Volatility is the central input to risk management, option pricing, and portfolio construction, and forecasting it one day ahead is a canonical problem in financial econometrics. The dominant practical model, the heterogeneous autoregressive model of realized volatility (HAR-RV) [2], is deceptively simple: it predicts tomorrow’s log realized variance as a linear combination of the most recent daily value, the trailing weekly average, and the trailing monthly average. Its effectiveness comes from an explicit *multi-timescale* inductive bias, the heterogeneous-market hypothesis that traders operating at different horizons each leave a footprint in volatility.

The motivation for this project is an observation that this inductive bias is not unique to finance. In recent co-authored work on human activity recognition (HAR) for ambient-assisted living [8], we proposed a hybrid temporal-convolutional LSTM that processes a short-context window (two minutes, capturing brief actions) and a long-context window (two hours, capturing extended routines) on two parallel branches before fusing them. The structural correspondence is exact: HAR-RV’s daily, weekly, and monthly aggregates are a fixed, hand-engineered version of the same short-versus-long multi-resolution decomposition that the deep model learns. This paper asks whether transferring that architecture, with its information set held equal to HAR-RV’s, improves volatility forecasts, and whether any improvement is statistically real rather than an artifact of a single training run. We frame prediction quality with both a statistical loss and a robustness analysis, and we are explicit about the limits imposed by the data.

## 2 Related Work and the HAR Analogy

HAR-RV [2] approximates the long-memory dynamics of realized variance with three lag aggregates and remains a strong baseline that newer models are routinely measured against. GARCH [1] models the conditional variance of returns directly and is the classical benchmark. On the deep-learning side, recurrent and convolutional models have been applied to volatility, but they rarely make their multi-timescale prior explicit.

Our prior HAR architecture [8] makes that prior its central design choice. It uses two parallel sliding windows, a two-minute short-term window and a two-hour long-term window, each processed by temporal convolutional layers with dilated convolutions that expand the receptive field without inflating the parameter count, and fuses the two branches through a recurrent step. Figure 1 places the two designs side by side: HAR-RV fixes the timescales and the linear weights a priori, whereas the dual-resolution TCNN-LSTM learns convolutional features over the same short and long horizons. This paper treats the deep model as a learned generalization of HAR-RV and tests whether the added flexibility pays off at the daily horizon.



Shared multi-timescale prior: fixed linear aggregation (HAR-RV) vs learned convolutional branches (TCNN-LSTM)

**Figure 1:** The shared multi-timescale prior. HAR-RV (left) combines fixed daily, weekly, and monthly aggregates linearly; the dual-resolution TCNN-LSTM (right) learns convolutional features over a short and a long window before fusing them. The transfer holds the information set equal and replaces the fixed linear aggregation with a learned nonlinear one.

## 3 Proposed Method

**Forecasting target.** All models forecast next-day log realized variance,  $y_t = \log RV_t$ , which stabilizes variance and approximate normality.

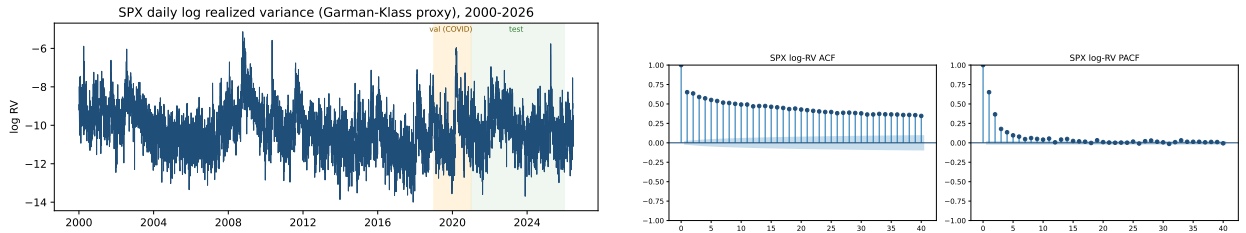
**Baselines.** HAR-RV regresses  $y_{t+1}$  on  $y_t$ , the trailing five-day mean, and the trailing 22-day mean by ordinary least squares. GARCH(1,1) is fitted on daily returns; because it models close-to-close return variance rather than the intraday range proxy used as the target, its conditional-variance forecasts are aligned to the proxy level by a single additive constant in log space estimated *only* on the training set, which removes a definitional level offset without using any validation or test information.

**Transferred model.** The dual-resolution TCNN-LSTM consumes a short window of five

trading days and a long window of 22 trading days, chosen so the combined information set matches HAR-RV’s daily-to-monthly span. Each window is processed by a stack of dilated 1-D convolutions (kernel 3, dilations 1, 2, 4) followed by global average pooling; the two pooled feature vectors are concatenated and passed as a length-one sequence through a single LSTM step and a small MLP head that emits the scalar forecast. This mirrors the published HAR model [8], with the input reduced to a single channel (univariate log-RV) and the objective changed from classification to mean-squared-error regression. The network has 89,153 parameters and is trained with Adam (learning rate  $10^{-3}$ , cosine schedule), batch size 64, and early stopping on validation RMSE.

## 4 Data and Implementation

The Oxford-Man Institute Realized Library, the standard source of 5-minute realized variance, was discontinued in 2022, so we construct a daily realized-variance proxy from S&P 500 index OHLC data retrieved through Yahoo Finance (2000–2026). For each day we compute the Garman-Klass range-based variance estimator [4] and model its logarithm. The sample is split chronologically into training (2000–2018, 4779 days), validation (2019–2020, 505 days), and test (2021–2025, 1255 days), with the validation window deliberately spanning the COVID-19 shock as an out-of-sample stress test. The log-RV series shows the canonical features of volatility (Figure 2): pronounced clustering around 2008 and 2020, mild positive skew, and a slowly decaying autocorrelation function with a partial autocorrelation that cuts off after roughly two lags, the long-memory signature that motivates the HAR decomposition. We emphasize that the target is a daily *range-based proxy*, not an intraday realized measure, which has direct consequences for the achievable accuracy (Section 6).



**Figure 2:** SPX daily log realized variance (left; validation and test windows shaded) and its autocorrelation and partial autocorrelation (right). The slow ACF decay with a sharp PACF cutoff is the long-memory signature underlying HAR-RV.

All experiments are tracked with MLflow on a SQLite backend, with seeds fixed and logged. The stack is Python 3.13 with PyTorch [6] for the neural model, the `arch` package for GARCH, and `statsmodels` for HAR-RV. The full environment is pinned and the pipeline is regenerable from a single entry point; code is available in the project repository (see Appendix).

## 5 Results

Table 1 reports forecast accuracy. On the test set the transferred TCNN-LSTM attains the lowest squared-error loss (log-RMSE 0.823) ahead of HAR-RV (0.835) and GARCH (0.884). Diebold-Mariano tests [3] with the small-sample correction of Harvey et al. [5] confirm the ordering under squared-error loss is statistically significant (Table 2): the TCNN-LSTM improves on HAR-RV

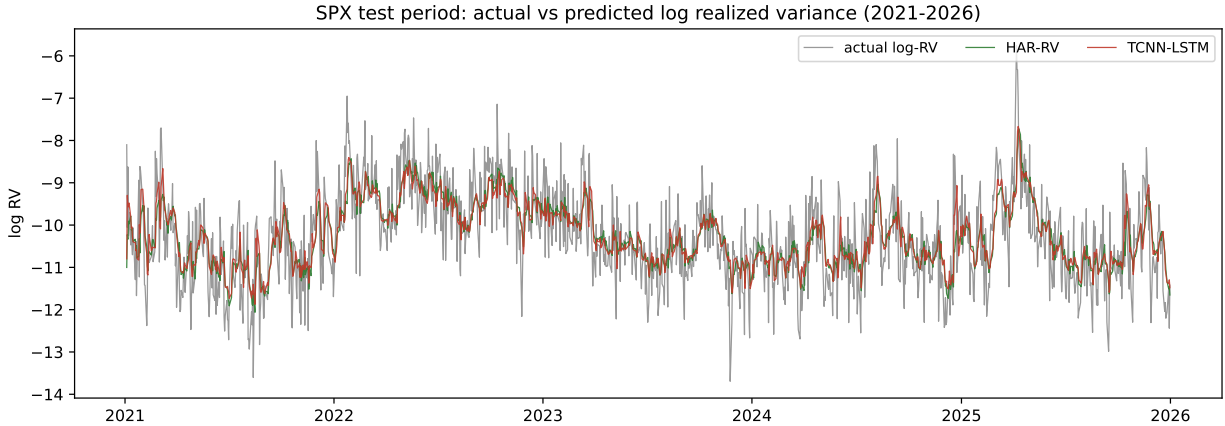
( $p = 0.013$ ) and on GARCH ( $p < 0.001$ ), and HAR-RV improves on GARCH ( $p < 0.001$ ). Figure 3 shows that all models track the test-period dynamics, with the deep model following the peaks slightly more closely.

**Table 1:** Forecast accuracy on SPX. RMSE is on log-RV; QLIKE [7] is on RV. Lower is better; best test values in bold.

| Model      | Val log-RMSE | Val QLIKE | Test log-RMSE | Test QLIKE   |
|------------|--------------|-----------|---------------|--------------|
| GARCH(1,1) | 0.947        | 0.457     | 0.884         | 0.454        |
| HAR-RV     | 0.896        | 0.515     | 0.835         | 0.445        |
| TCNN-LSTM  | 0.879        | 0.471     | <b>0.823</b>  | <b>0.424</b> |

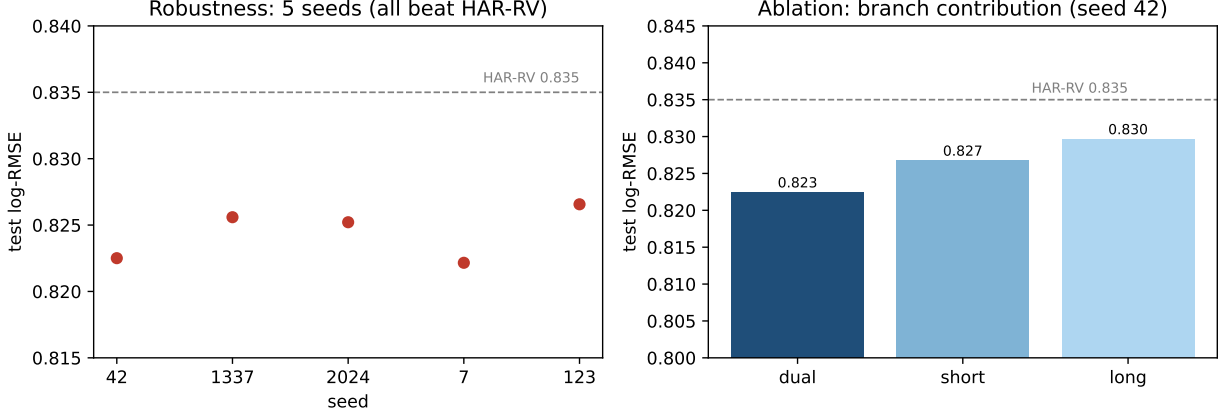
**Table 2:** Pairwise Diebold-Mariano tests on the test set (negative statistic favors the first model). The RMSE improvements are significant; under QLIKE only the HAR-RV vs TCNN-LSTM pair is significant.

| Pair                | SE stat | SE $p$ | QLIKE stat | QLIKE $p$ |
|---------------------|---------|--------|------------|-----------|
| GARCH vs HAR-RV     | +4.71   | 0.000  | +0.47      | 0.638     |
| GARCH vs TCNN-LSTM  | +5.38   | 0.000  | +1.27      | 0.203     |
| HAR-RV vs TCNN-LSTM | +2.49   | 0.013  | +2.17      | 0.030     |



**Figure 3:** Actual versus predicted log realized variance over the SPX test period (2021–2026).

**Robustness and ablation.** Because the headline comparison used a single random seed, we trained the model under five seeds and ran a branch ablation (Figure 4). The squared-error improvement is robust: all five seeds beat HAR-RV with a test log-RMSE of  $0.824 \pm 0.002$ . The QLIKE advantage is not robust; only one of five seeds beats HAR-RV on QLIKE, so we report the squared-error result as the finding and treat the seed-42 QLIKE win as indicative only. The ablation shows the dual-resolution design contributes: removing the short branch raises test RMSE to 0.827 and removing the long branch to 0.830, so both timescales carry signal, with recent history dominant.



**Figure 4:** Left: five-seed robustness; every seed beats HAR-RV on test RMSE. Right: branch ablation; the dual model outperforms either single branch.

## 6 Discussion and Limitations

The transfer succeeds in a measurable but modest way: replacing HAR-RV’s fixed linear aggregation with learned convolutional branches over the same information set yields a small, robust, and statistically significant reduction in squared forecast error, while leaving variance-space (QLIKE) loss essentially unchanged. Three limitations frame this result honestly. First, the absolute RMSE of every model is high relative to the realized-volatility literature because our target is a daily Garman-Klass proxy rather than a 5-minute realized measure; this measurement noise is irreducible and bounds the achievable error for all models equally, so only the relative comparison is informative, and it likely compresses the gap between a linear and a nonlinear model. Second, the study covers a single asset (SPX) and a single architecture trained once per seed; broader cross-sectional and multi-architecture evidence would strengthen the claim. Third, the GARCH level calibration, while estimated only on training data, is a modeling choice that a stricter evaluation might replace with a model matched to the same proxy.

The most useful reading of the result is about where the multi-timescale prior pays off. At the daily horizon with a noisy proxy, Corsi’s linear decomposition is already near-optimal, so the learned model gains little. The architecture’s value in the HAR domain came from genuinely nonlinear interactions among sensor channels; financial volatility at this horizon may simply not contain enough nonlinearity to reward the inductive bias. This points to intraday horizons, regime-switching periods, and multi-asset cross-sections, where nonlinearity is stronger, as the more promising settings for the transfer.

## 7 Conclusion

We transferred a dual-resolution hybrid TCNN-LSTM from human activity recognition to daily volatility forecasting, motivated by the structural identity between its short-and-long-context design and HAR-RV’s multi-timescale decomposition. On SPX the transferred model significantly and robustly improves on HAR-RV and GARCH under squared-error loss, with the improvement validated across seeds and attributed to the dual-resolution structure by ablation, while being honestly bounded by the noise of the daily proxy. The contribution is a clean, testable bridge from a

known financial inductive bias to a learned architecture that respects it.

## References

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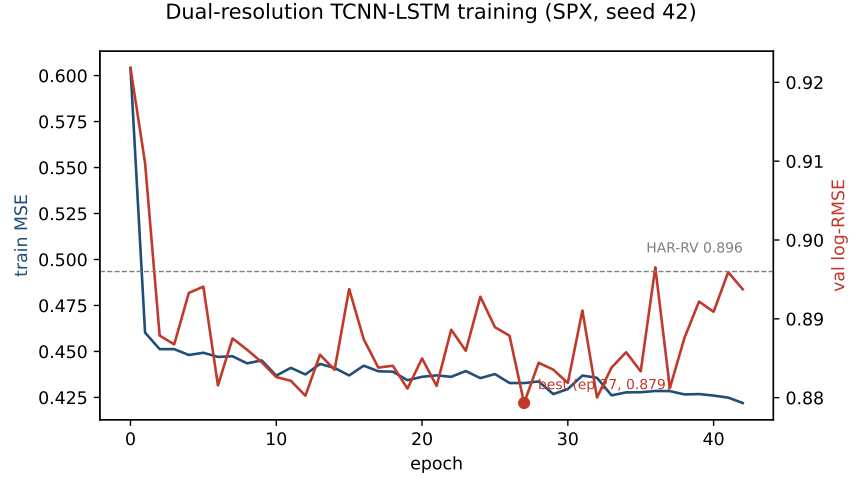
## A Hyperparameters and Fitted Coefficients

TCNN-LSTM: short window 5, long window 22, TCN channels (32, 64), dilations (1, 2, 4), LSTM hidden 64, dropout 0.1, Adam lr  $10^{-3}$  with cosine schedule, batch 64, max 200 epochs, early stopping patience 15, seed 42; 89,153 parameters; best epoch 27. HAR-RV coefficients (SPX): const  $-0.678$ , daily 0.149, weekly 0.462, monthly 0.323 (slopes sum 0.933, reproducing Corsi’s near-unit persistence). GARCH(1,1):  $\mu = 0.062$ ,  $\omega = 0.025$ ,  $\alpha = 0.120$ ,  $\beta = 0.861$  (persistence 0.981), log-level calibration offset  $-0.870$ .

## B Reproducibility and Code

All runs are logged to MLflow with fixed seeds. The repository contains the data pipeline, models, training loop, evaluation, and the robustness and ablation drivers, regenerable end to end. Repository: <https://github.com/MohamedJJ824/volatility-project>.

## C Training Curve



**Figure 5:** TCNN-LSTM training: train MSE and validation log-RMSE per epoch, with the early-stopped best epoch and the HAR-RV reference marked.

## D Use of AI Tools

Claude Code (claude-opus-4-8) was used as a coding and writing assistant throughout, under the course policy permitting AI coding tools with disclosure. It was used to scaffold the data pipeline, the GARCH/HAR-RV baselines, the TCNN-LSTM and training loop, the Diebold-Mariano test, the figures, and a draft of this report. All outputs were verified by the author: model coefficients were checked against known regularities (HAR slopes positive and summing near one; GARCH persistence near 0.98), the RMSE floor was confirmed by benchmarking against climatology and a random walk, the significant single-seed result was re-tested across five seeds, and every figure and number was inspected before inclusion. A cumulative tool-usage log is maintained in the repository.